Ex. 9.4 Q 24

$$\sum_{n=3}^{\infty} \frac{5n^3 - 3n}{n^2(n-2)(n^2+5)} \sim \frac{5n^3}{n^2 n n^2} = \boxed{\frac{5}{n^2}}$$

~~Df~~ Draft. $a_n = \ldots$

$b_n = \frac{5}{n^2}$ for $n \geq 3$

$2 > 1$

$\lim_{n\to\infty} \frac{a_n}{b_n} = \cdots = 1$ $\Rightarrow \sum \frac{5}{n^2}$ conv.

Since ......, by LCT, ...

Q49 $\displaystyle\sum_{n=1}^{\infty} \frac{\coth n}{n^2} \sim \frac{c}{n^2} \rightarrow$ Conv. (p-series)

$$\underline{\coth n} = \frac{\cosh n}{\sinh n} = \frac{(e^n + e^{-n})/2}{(e^n - e^{-n})/2}$$

Conv. seq.

$$= \frac{1 + e^{-2n}}{1 \cancel{+} - e^{-2n}} \rightarrow 1$$

$a_n = \cdots$

$b_n = \dfrac{1}{n^2}$

$\displaystyle\lim \frac{a_n}{b_n} = \cdots = 1$

[Conclusion]

Q51 $\sum \frac{1}{n\sqrt[n]{n}}$ $\sqrt[n]{n} \to 1$

conv. seq.

$\frac{1}{n\sqrt[n]{n}} \sim \frac{1}{n}$ (LCT with $b_n = \frac{1}{n}$)

Q52 $\sum \frac{\boxed{\sqrt[n]{n}}}{n^2} \sim \frac{1}{n^2}$ (LCT)

Ex. 9.5 Q46

$$\sum \frac{2^{n^2}}{n^{2^n}}$$

① $2^n \geqslant 2n^2 \Rightarrow 2^{2^n} \geqslant 2^{2n^2}$

$$2^{2^n} \leqslant n^{2^n}$$

$$\Rightarrow \frac{2^{n^2}}{n^{2^n}} \leqslant \frac{2^{n^2}}{2^{2n^2}} = \frac{1}{2^{n^2}} \leqslant \frac{1}{2^n} = \left(\frac{1}{2}\right)^n$$

② (Equalize the powers)

$$3^{n^2} \leq n^{n^2} \leq \underline{\underline{n^{2^n}}}$$

$$\Rightarrow \frac{1}{\underline{n^{2^n}}} \leq \frac{1}{\underline{3^{n^2}}} \Rightarrow \frac{2^{n^2}}{n^{2^n}} \leq \left(\frac{2}{3}\right)^{n^2}$$

③ Root Test (Tedious)